**Credit Card Analytics Dashboard – Overview**
**Project Description**
This project focuses on building an end-to-end **Credit Card Analytics Dashboard** using Power BI to analyze customer behavior and transaction trends. The solution consists of two interactive dashboards: a **Customer Report** and a **Transaction Report**, designed to provide actionable insights for business decision-making.
The dashboards enable stakeholders to explore patterns in revenue, customer segmentation, and transaction activity through dynamic filters and visually intuitive reports.
The overall objective aligns with the requirement of creating **interactive, insight-driven dashboards using customer and transaction datasets** .

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**Approach & Methodology**
The project was executed in a structured data analytics workflow:
**1. Data Collection & Integration**

- Imported two datasets into Power BI:
    - **Customer Data** (demographics, income, gender, etc.)
    - **Transaction Data** (date, amount, category, etc.)
- Established a relationship using **Client_Num** to enable cross-table analysis.

**2. Data Cleaning & Transformation**

- Used **Power Query Editor** to:
    - Handle missing values and inconsistencies
    - Standardize data formats (especially date fields for weekly analysis)
    - Remove duplicates and ensure data integrity

**3. Data Modeling**

- Created a **relational data model** connecting customer and transaction tables.
- Ensured proper cardinality and filter direction for accurate reporting.

**4. Feature Engineering (DAX)**
Developed calculated columns and measures such as:

- **Age Group** (e.g., 18–25, 26–35)
- **Income Group** (Low, Medium, High)
- **Total Revenue**
- **Current Week Revenue**
- **Previous Week Revenue**
- **Revenue Growth %**
- **Week Number**

These calculations enabled **time-based analysis and customer segmentation** .
**5. Dashboard Development**
Built two interactive dashboards:
**Dashboard 1: Customer Report**

- Focus: Customer demographics & segmentation
- Filters:
    - Gender
    - Age Group
    - Income Group
- Visuals:
    - Customer distribution
    - Revenue contribution by segment
    - KPI cards

**Dashboard 2: Transaction Report**

- Focus: Transaction patterns & revenue trends
- Filters:
  - Gender
  - Week
  - Spending Category
- Visuals:
  - Weekly revenue trends
  - Transaction volume
  - Category-wise spending

**6. Interactivity & UX**

- Implemented:
  - **Slicers & filters**
  - **Drill-down capability**
  - **Dynamic KPI cards**
- Ensured dashboards are:
  - Clean
  - Intuitive
  - Business-friendly

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**Key Insights**

**Customer Insights**

- The dataset includes **~10.29K customers**, with:
  - Average income: **57.09K**
  - Average satisfaction score: **3.19** (moderate satisfaction)
- **Female customers (58.17%)** contribute significantly higher revenue (**1.75M**) compared to males (**1.25M**), indicating stronger engagement.
- The **medium-income group (3.8K customers)** forms the largest segment, followed by:
  - Low income: 3.4K
  - High income: 3.0K
    → Suggests a strong middle-class customer base.
- **Top revenue-generating occupations:**
  - Self-employed: **0.76M**
  - Businessmen: **0.57M**

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**Transaction Insights**

- Total revenue is approximately **3M**, indicating strong financial activity.
- Weekly performance shows:
  - **Current week:** 51K
  - **Previous week:** 47K
  - **Growth: +10.15%** → positive upward trend
- Around **10K transactions** recorded → high engagement level.

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**Product Insights**

- Revenue is highly concentrated in:
  - **Blue Card:** 2.73M
- Other categories contribute significantly less:
  - Silver: 0.19M

    - Gold: 0.06M
    - Platinum: 0.02M

→ Indicates **product imbalance and potential upselling opportunity**.

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**Challenges Faced**

**1. Data Quality Issues**

- Inconsistent data types (especially date fields)
- Required careful preprocessing for accurate weekly analysis

**2. Data Modeling Complexity**

- Ensuring correct relationship between datasets
- Avoiding ambiguity in filtering and aggregation

**3. DAX Calculations**

- Creating time-based metrics like:
    - Week-over-week revenue
    - Growth percentage
- Required proper context handling

**4. Dashboard Design Balance**

- Maintaining clarity while adding interactivity
- Avoiding clutter while presenting multiple KPIs

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**Conclusion**

The project successfully delivers an **interactive and insight-driven analytics solution** that enables stakeholders to:

- Understand customer behavior
- Track revenue trends
- Identify high-value segments
- Make data-driven business decisions